



Habib University

shaping futures

Course Title: **LINEAR ALGEBRA**

Instructors:

Aeyaz J Kayani, Sajal S Rana

Class ID: **MATH 205**

Total Marks: **100**

Duration: **3 hours**

	Marks Obtained	Instructions for Students: a. Read the following instructions carefully. b. Please ensure that your mobile phone is switched off. c. All your personal belongings must be placed at the designated area of the examination hall. d. Write your Name, ID, Subject, Section, Instructor's name, and date in the space provided below. e. Remember to sign the exam attendance sheet circulated by the head invigilator / proctor. f. You may use an A4 single-sided handwritten formula sheet, which must be submitted with the rest of your work at the end of the exam. g. Answer all questions in the exam. h. You must provide complete working and reasoning for all steps of your solution. i. If you need to leave the room for any reason, please raise your hand and take permission from head invigilator / proctor. j. After finishing your exam, raise your hand and give your answer book to the head invigilator / proctor.
Q. 1		
Q. 2		
Q. 3		
Q. 4		
Q. 5		
Q. 6		
Q. 7		
Q. 8		
Q. 9		
Q. 10		
Total		
Instructor's Signature & Date		

Name: _____

Student ID: _____ Section: _____

Subject: _____

Instructor: _____

Date: _____



Question 1

- (a) Let $\underline{u}$ and $\underline{v}$ be vectors in the Euclidean space $\mathbb{R}^k$. For the Euclidean inner product, show that $\underline{u} \cdot \underline{v} = \frac{1}{4}(\|\underline{u} + \underline{v}\|^2 - \|\underline{u} - \underline{v}\|^2)$ [5]

- (b) For a Euclidean space $\mathbb{R}^n$ show that $d(\underline{u}, \underline{v}) \leq d(\underline{u}, \underline{w}) + d(\underline{w}, \underline{v})$, for all $\underline{u}, \underline{v}$ and $\underline{w}$ in $\mathbb{R}^n$. [5]



Question 2

(a) Let S be the set of all real numbers in $(0, \infty)$. If S is a vector space under the operations $\underline{u} + \underline{v} = \underline{u \cdot v}$ and $k\underline{u} = \underline{u^k}$, then identify the additive identity $\underline{0}$ of S and the additive inverse $-\underline{u}$. Hence, show that Axioms 4 and 5 for vector spaces are satisfied for S . **[6]**

(b) A set with exactly two vectors is linearly independent if and only if neither vector is a scalar multiple of the other **[4]**



Question 3

Let V be a vector space and let $\underline{u}$ be an element of V . Let k be a scalar. Show that:

(a) $k\underline{0} = \underline{0}$ [5]

(b) $-1\underline{u} = -\underline{u}$ [5]



Question 4

Let S be a finite set of vectors in a finite-dimensional vector space V . Fill in the blanks in the statements of the theorem below, and then prove both.

- (a) If S spans V but is not a basis for V , then S can be _____ to a basis for V by _____ appropriate vectors from S . **[5]**

- (b) If S is a linearly independent set that is not already a basis for V , then S can be _____ to a basis for V by _____ appropriate vectors into S . **[5]**



Question 5

Prove: If A is an $n \times n$ diagonalizable matrix, then A has n linearly independent eigenvectors. **[10]**



Question 6

(a) Consider the basis $S = \{\underline{v}_1, \underline{v}_2, \underline{v}_3\}$ for $\mathbb{R}^3$, where $\underline{v}_1 = (1, 1, 1)$, $\underline{v}_2 = (1, 1, 0)$, and $\underline{v}_3 = (1, 0, 0)$. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the linear transformation such that $T(\underline{v}_1) = (1, 0)$, $T(\underline{v}_2) = (2, -1)$, $T(\underline{v}_3) = (4, 3)$. Find a formula for $T(x_1, x_2, x_3)$. [5]

(a) Use your formula from part (a) to find $T(1, -5, 0)$. [2]

(c) Find the matrix of the linear transformation in the parts above. [3]



Question 7

Let $f(x)$ and $g(x)$ be continuous functions on $[0, 1]$. Prove that:

[10]

$$\left[\int_0^1 f(x) g(x) dx \right]^2 \leq \left[\int_0^1 f^2(x) dx \right] \left[\int_0^1 g^2(x) dx \right]$$



Question 8

If $\underline{v_1}, \underline{v_2}, \dots, \underline{v_r}$ are vectors in a vector space U , then prove the following:

(a) The set W of all linear combinations of $\underline{v_1}, \underline{v_2}, \dots, \underline{v_r}$ is a subspace of U . [5]

(b) W is the smallest subspace of U that contains $\underline{v_1}, \underline{v_2}, \dots, \underline{v_r}$. [5]



Question 9

For each of the following statements, decide if they are true or false, along with a reason:

- (a) If A is any $m \times n$ matrix, then $A^T A$ has an orthonormal set of n eigenvectors. [3]

- (b) If A is a diagonalizable matrix then the rank of A is the number of nonzero (may not be distinct) eigenvalues of A . [5]

- (b) If A is a diagonalizable $n \times n$ matrix, then A has n distinct eigenvalues. [2]



Question 10

Let $T: P_2 \rightarrow P_3$ be the linear transformation given by $T(p(x)) = x p(x)$.

(a) Which of x , 0 , and $(1 - x)$ are in $\ker(T)$? [3]

(b) Is T an onto transformation? Why or why not? [4]

(c) What is the dimension of $\ker(T)$? Can you suggest a basis for it? [3]



ROUGH WORK



ROUGH WORK